21_Data_Processing

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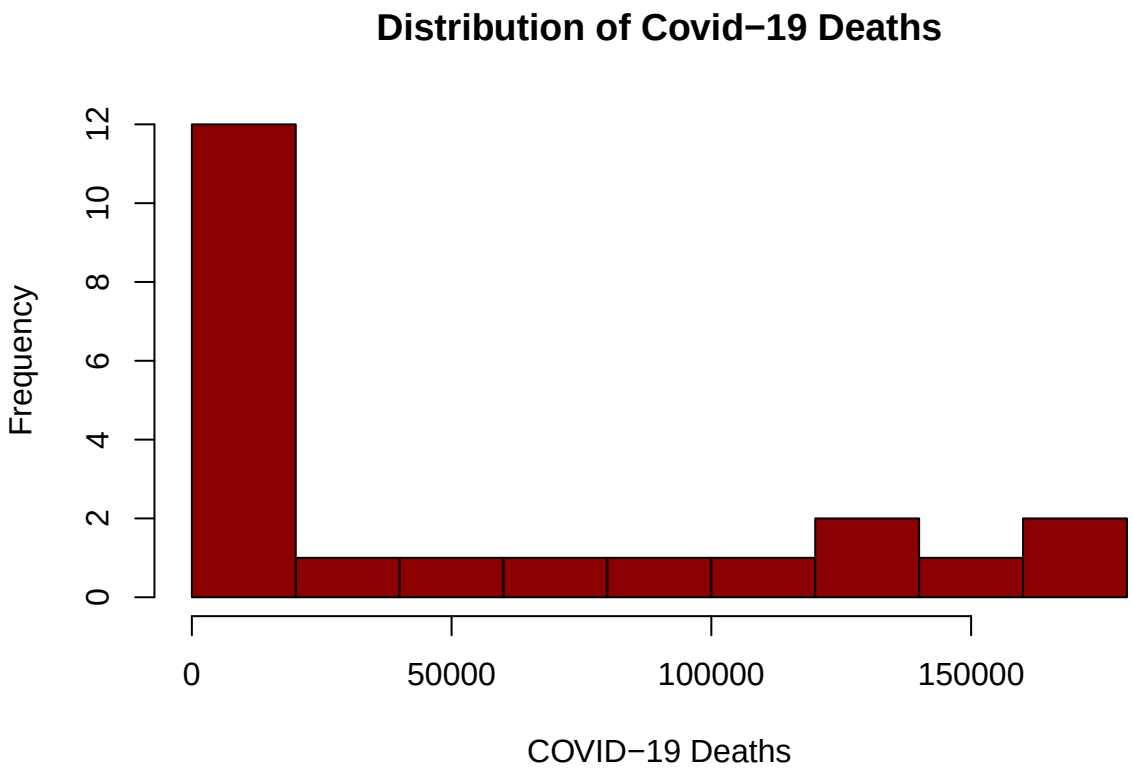
Loading the Clean Dataset

```
load("12_DataSaving.RData")
head(covid_cleaned)
```

```
##     Sex    Age.Group COVID.19.Deaths Total.Deaths Pneumonia.Deaths
## 19 Male Under 1 year             284        40371              586
## 21 Male    1-4 years             151         8102              341
## 22 Male   5-14 years             261        12850              449
## 23 Male  15-24 years            1797        97748             1917
## 25 Male  25-34 years            7607       196824             7129
## 27 Male  35-44 years           18602       273868            16415
##    Pneumonia.and.COVID.19.Deaths
## 19                            54
## 21                            35
## 22                            83
## 23                           730
## 25                          3594
## 27                          9399
```

Plot 1: Histogram of COVID-19 Death Distribution

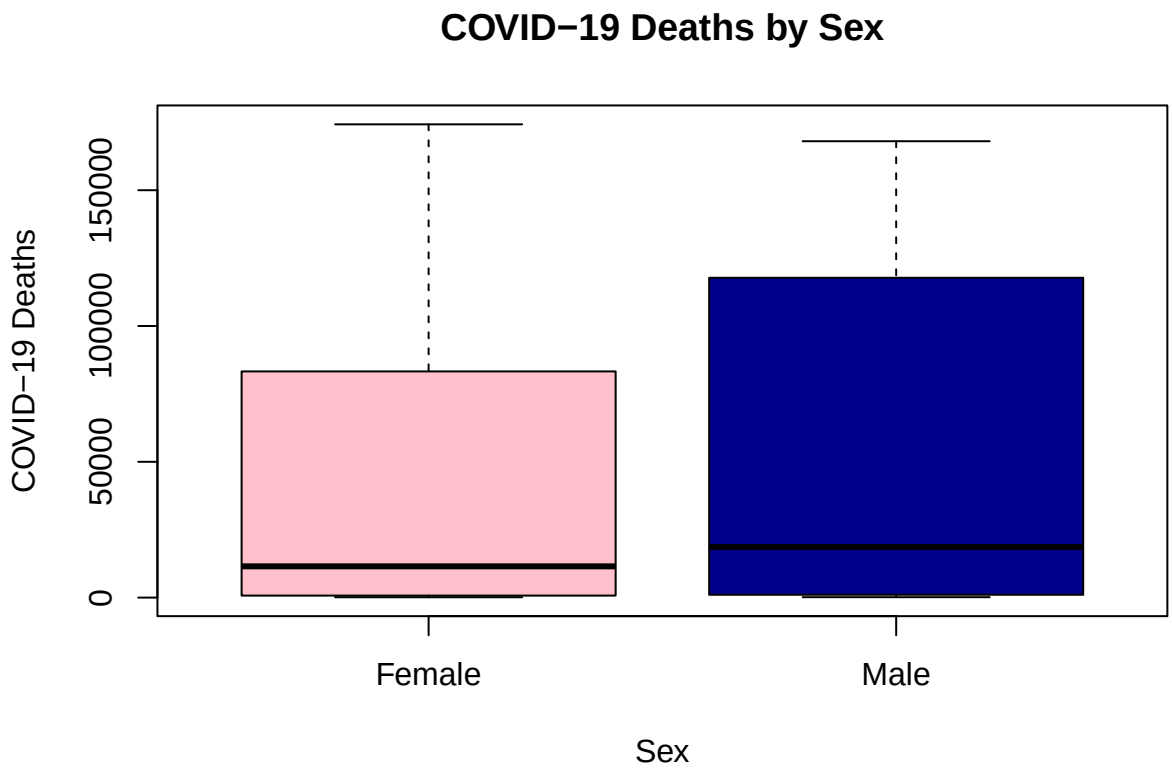
```
hist(covid_cleaned$COVID.19.Deaths,
    main = "Distribution of Covid-19 Deaths",
    xlab = "COVID-19 Deaths",
    col = 'darkred',
    breaks = 10)
```



This histogram displays the overall distribution and frequency of COVID-19 Deaths counts across all filters. It highlights that the lower death counts show up as small parts whereas the high deaths show up as a outlier.

Plot 2: Boxplot of COVID-19 Deaths by Sex

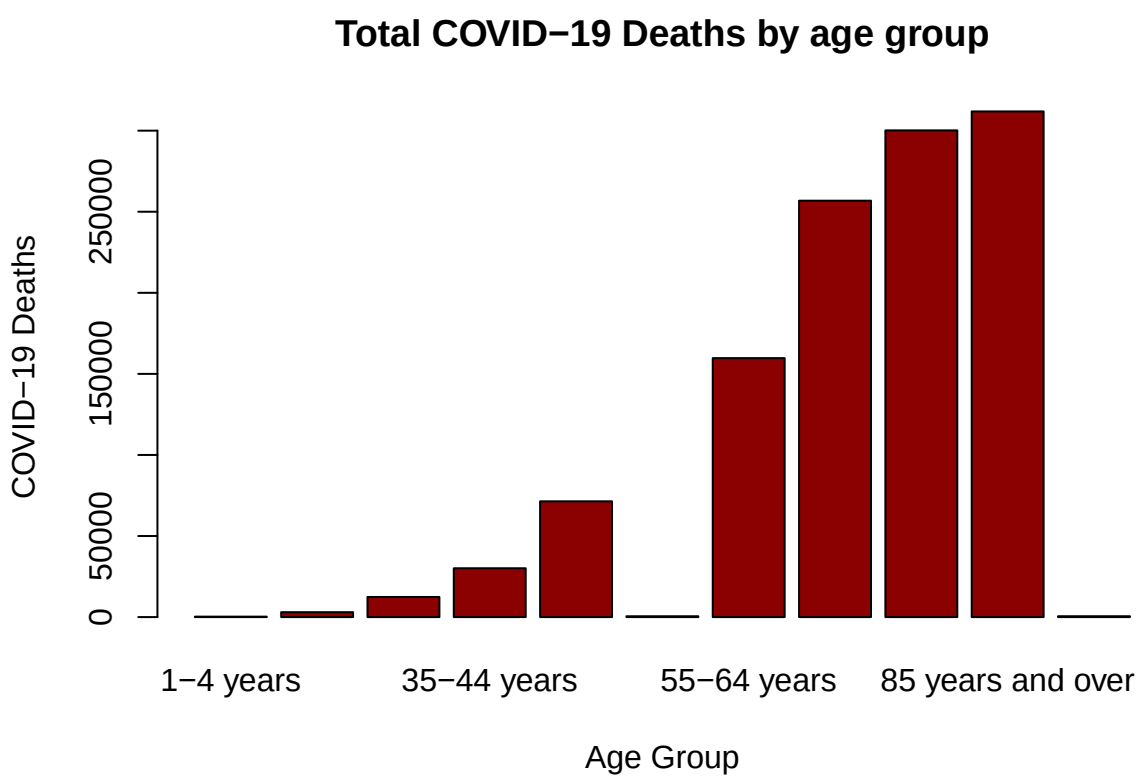
```
boxplot(COVID.19.Deaths ~Sex,
        data = covid_cleaned,
        main = "COVID-19 Deaths by Sex",
        xlab = "Sex",
        ylab = 'COVID-19 Deaths',
        col = c('pink','darkblue'))
```



This box plot compares the median, spread and range of COVID-19 deaths between males and females. It shows variability and highlights whether one group had a higher mortality rate

Plot 3: Bar Plot of COVID-19 deaths by age

```
age_totals <- tapply(covid_cleaned$COVID.19.Deaths, covid_cleaned$Age.Group, sum)
barplot(age_totals,
        main = 'Total COVID-19 Deaths by age group',
        xlab = 'Age Group',
        ylab = 'COVID-19 Deaths',
        col = 'darkred')
```



This bar plot shows the total COVID-19 Deaths across age groups only. We are clearly able to intrepret that as someone gets older, they have a higher mortality rate and are more prone to be affected by COVID-19

Plot 4: Scatter plot for percentage of all deaths

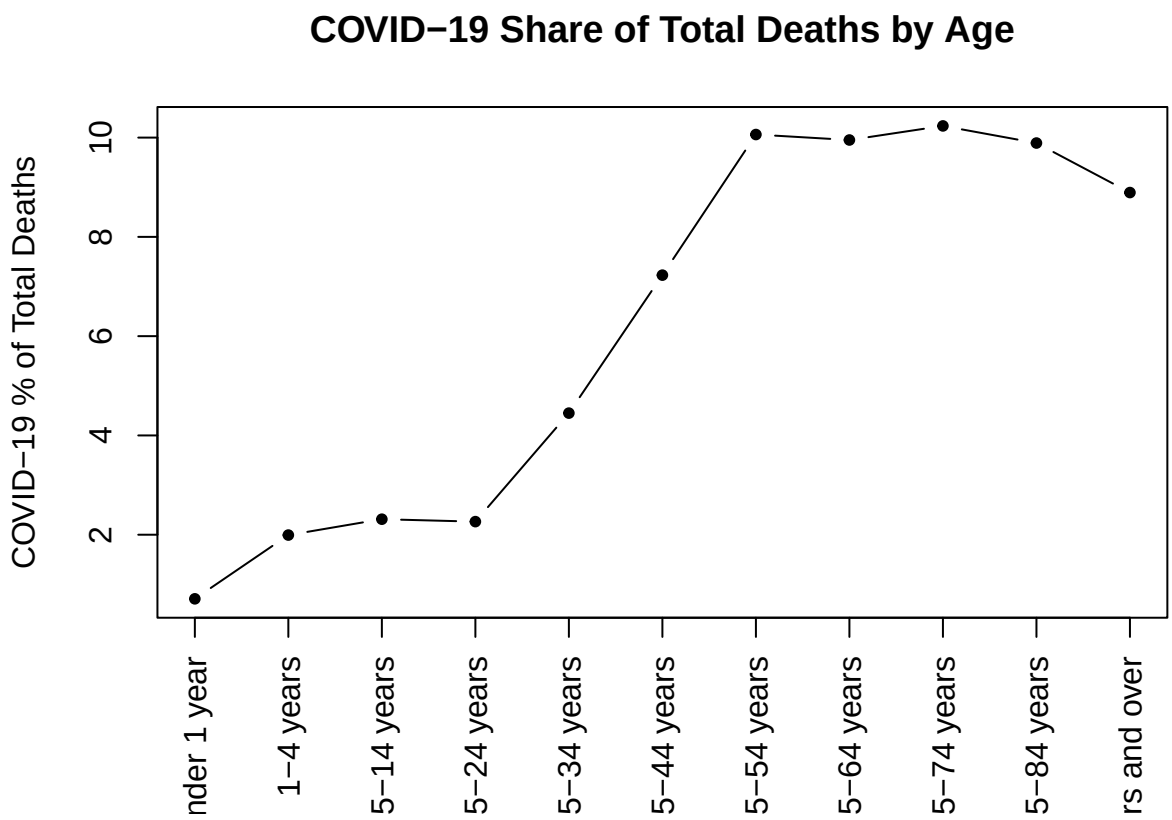
```
age_combined <- aggregate(
  cbind(COVID.19.Deaths, Total.Deaths) ~ Age.Group,
  data = covid_cleaned,
  sum
)

age_combined$Age.Group <- factor(
  age_combined$Age.Group,
  levels = c("Under 1 year", "1-4 years", "5-14 years", "15-24 years",
             "25-34 years", "35-44 years", "45-54 years", "55-64 years",
             "65-74 years", "75-84 years", "85 years and over"),
  ordered = TRUE
)
age_combined <- age_combined[order(age_combined$Age.Group), ]


pct <- (age_combined$COVID.19.Deaths / age_combined$Total.Deaths) * 100
```

```r
plot(
  pct,
  type = "b",
  pch = 20,
  xaxt = "n",
  xlab = '',
  ylab = "COVID-19 % of Total Deaths",
  main = "COVID-19 Share of Total Deaths by Age"
)

axis(1, at = 1:nrow(age_combined), labels = age_combined$Age.Group, las = 2)
```



This line plot shows the percentage of COVID-19 deaths relative to total deaths across the age groups. It portrays how the percentage of deaths change with the age groups.